

Are LLMs Optimal Information Processors?

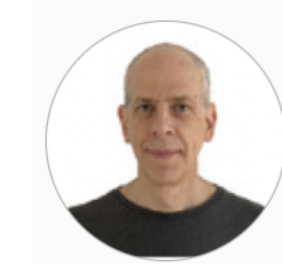
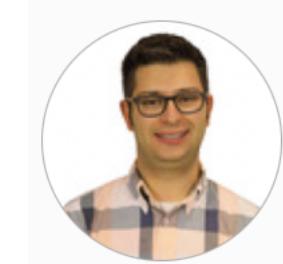
Residency Midpoint Presentation

Chacha Chen (*PAIS*)

02/18/2026

Intro

Mentors: Nick Foti, Guillermo Sapiro



PAIS team (Personalized AI for Science)

Project 1 (Research)

Are LLMs Optimal Information Processors?

Project 2 (Ultra Disclosed Product)

Focus area: using LLMs in the health space

How *good* are LLMs at *reasoning from empirical evidence*?

Example:



Vomiting —> “Cancer”

Over-update

How *good* are LLMs at *reasoning from empirical evidence*?

Example:



Overreacting to weak evidence

Vomiting —> “Cancer”

Over-update

How *good* are LLMs at *reasoning from empirical evidence*?

Example:



Lab tests —> “just infection”

Under-update

How *good* are LLMs at *reasoning from empirical evidence*?

Example:



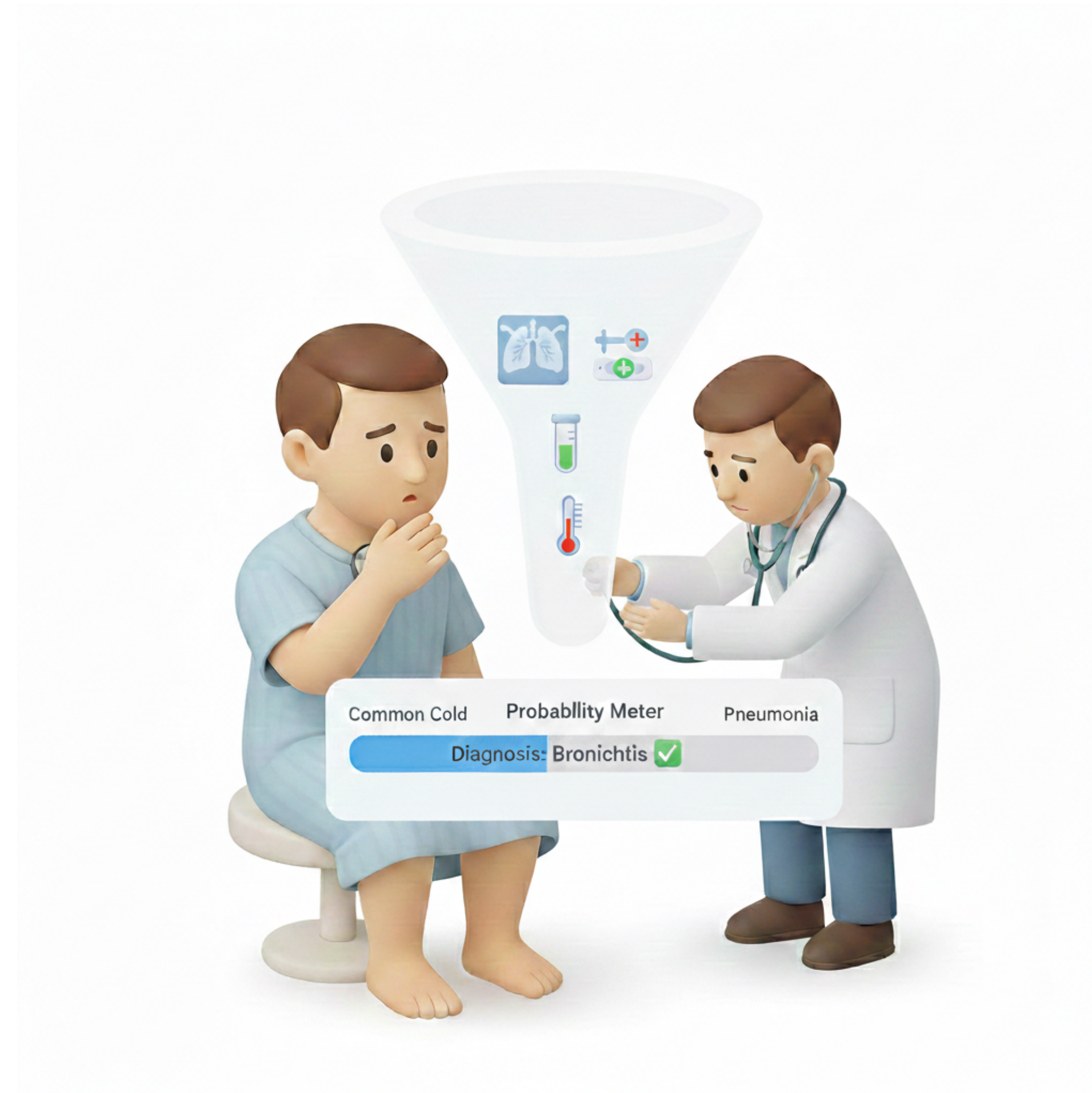
Ignoring strong
evidence

Lab tests —> “just infection”

Under-update

Key intuition for good reasoning: *We want to measure whether LLMs update its belief in **proportion** to evidence strength.*

Example:



"Good" update

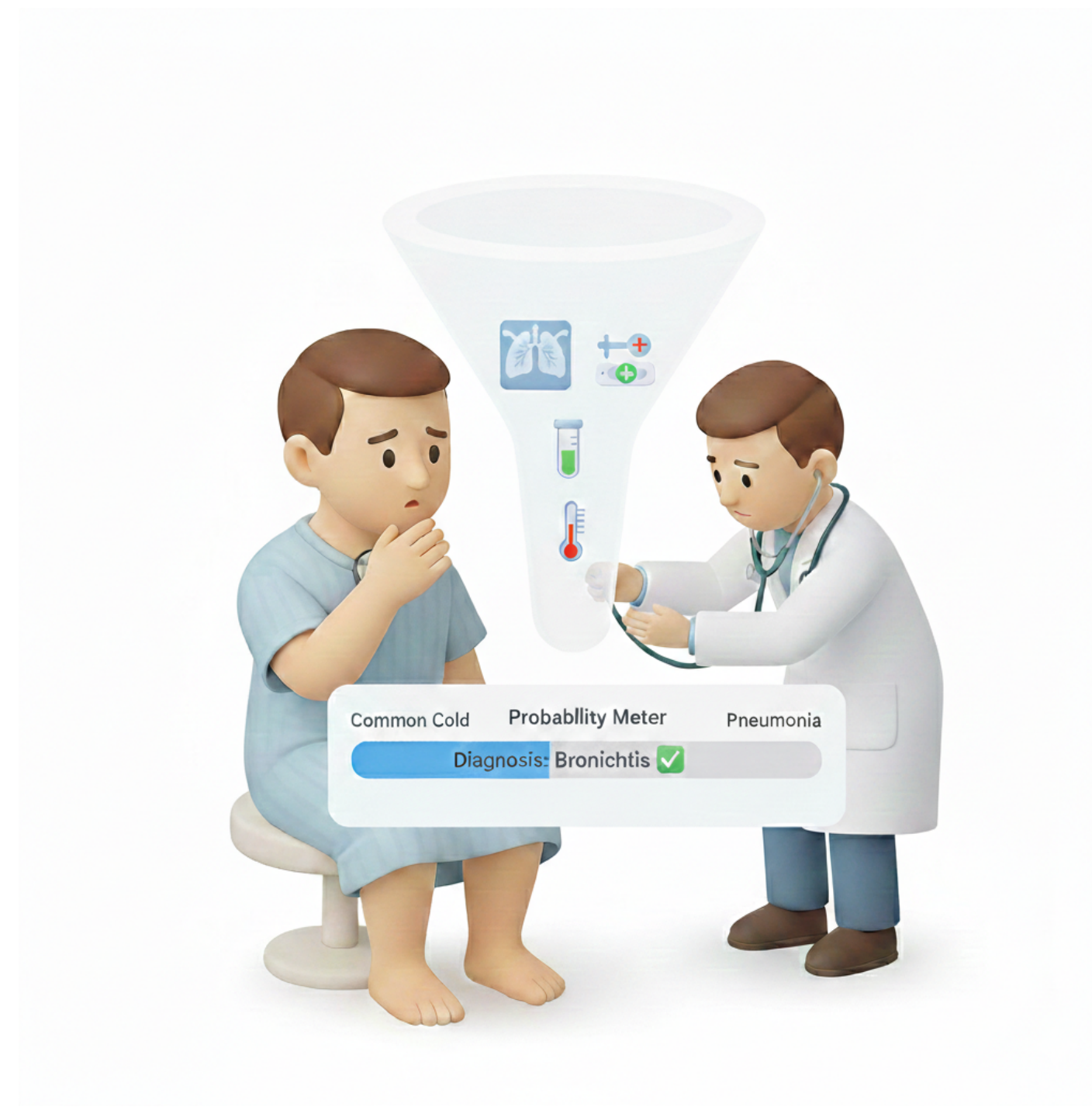
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Key intuition for measurement: *LLMs should update its belief in **proportion** to evidence strength.*



Vomit —> “Cancer”

Over-update



“Good” update



Lab tests —> “just infection”

Under-update

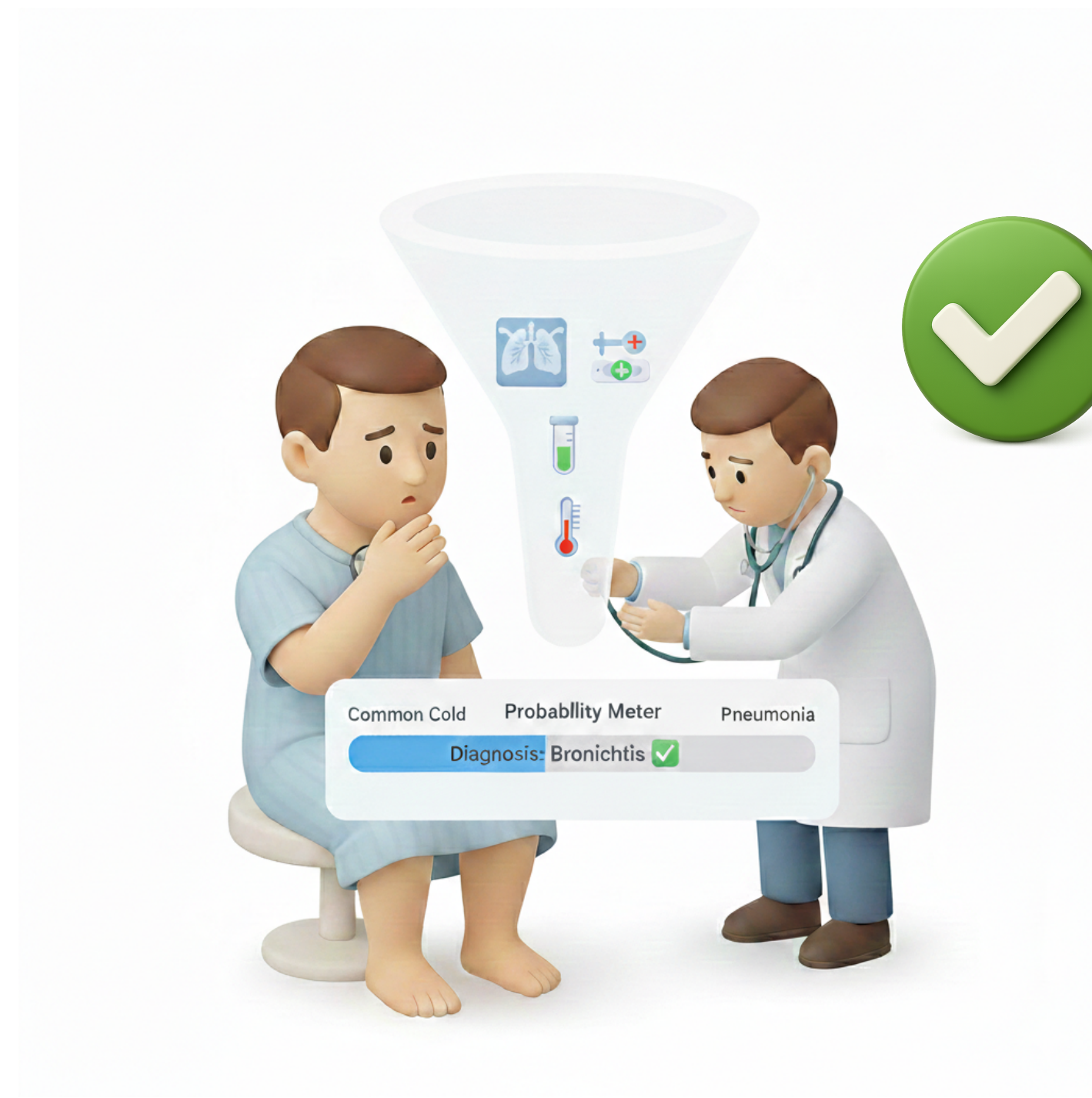
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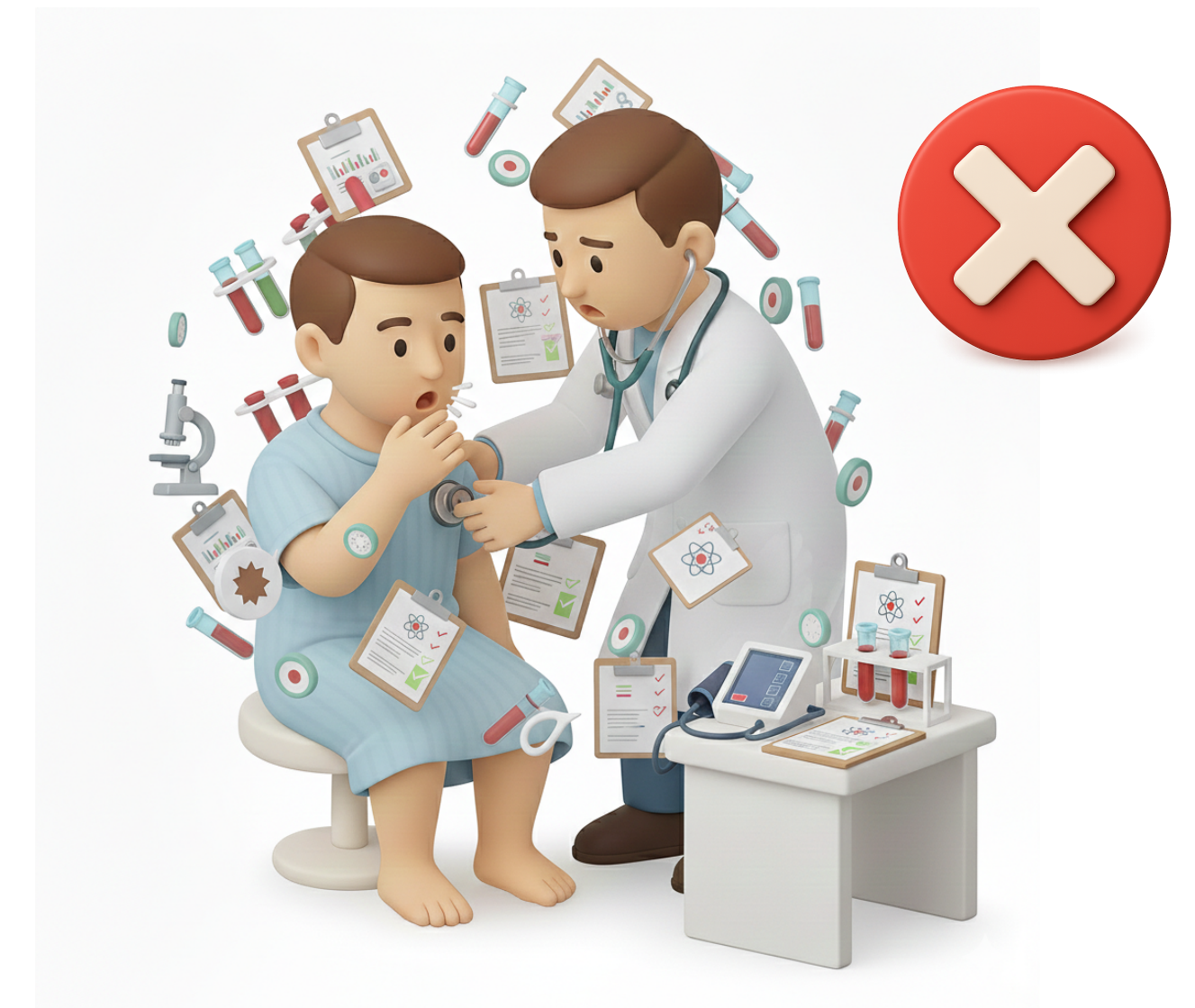


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“Good” update



Lab tests —> “just infection”

Under-update

Preliminary Findings/Progress

Model	Accuracy			Zellner Abs Delta		
	GPQA	MediQ	MedPair	GPQA	MediQ	MedPair
Closed Source						
gpt-4o-mini	0.412	0.806	0.596	0.208	0.188	0.183
gpt-5.1	0.740	0.933	0.860	0.215	0.134	0.168
Open Source						
DeepSeek-R1-Qwen3-8B	0.562	0.814	0.580	0.204	0.098	0.100
Qwen3-8B	0.458	0.713	0.598	0.331	0.231	0.232
Qwen3-14B	0.534	0.767	0.600	0.168	0.149	0.158
Ministral-3-8B-Instruct	0.414	0.679	0.560	0.298	0.227	0.219
Ministral-3-14B-Instruct	0.438	0.678	0.605	0.325	0.250	0.232
Ministral-3-8B-Reasoning	0.524	0.747	0.592	0.283	0.177	0.169
Ministral-3-14B-Reasoning	0.511	0.791	0.645	0.256	0.160	0.175

Zellner’s delta as a metric for reasoning quality: the lower the better

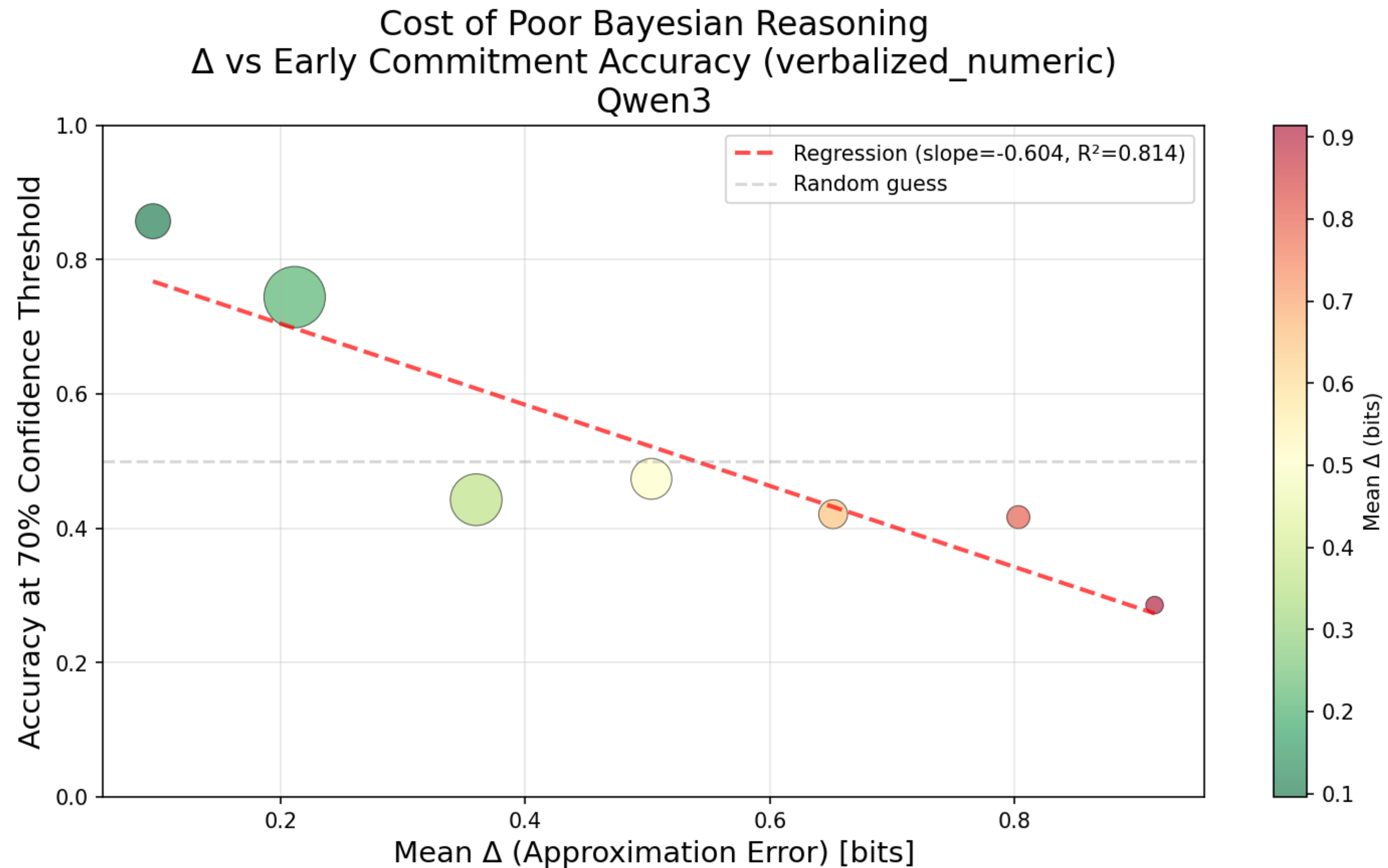
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Zellner’s delta as a metric for reasoning quality: the lower the better

Despite high final/reasonable accuracy, models show **large reasoning gaps**

Practical Consequences



Poor reasoning quality isn't just a theoretical issue—it causes real errors when models must decide under uncertainty

Key Takeaways

Our Findings

- **LLMs reasoning is not “optimal”**, they exhibit imperfect reasoning failures despite high accuracy.
- We propose a ***diagnostic metric***, Zellner's Delta, which evaluates how well those models reason from empirical evidence under uncertainty, providing more rich signals about model reasoning than accuracy metrics.

Future Directions: *Training* to improve reasoning quality

Language modeling objective + Zellner's Delta regularizer

$$\mathcal{L} = \underbrace{- \sum_t \log P(x_t | x_{<t})}_{\mathcal{L}_{LM}} + \lambda \cdot |\Delta|$$

where $\mathcal{L}_{LM}$ is next-token prediction on the full completion. The model learns to generate text. ICP regularizes the implicit probabilities (extracted from logits at the answer position) to be Bayesian-coherent.

